

Master Exam Spring 2019

Problem 1. Let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x \leq \infty.$$

- (1) Find a sufficient statistic for θ .
- (2) Find the MLE and the method of moments estimator of θ .

Solution. (1) Since $f(\mathbf{X}|\theta) = \theta^n (\prod_i X_i^{-2}) \cdot \mathbf{1}_{[\theta, \infty)}(X_{(1)})$ where $X_{(1)} < \dots < X_{(n)}$ is the order statistic, $X_{(1)}$ is a sufficient statistic.

(2) To maximize the log likelihood function $L(\theta|\mathbf{X}) = \log f(\mathbf{X}|\theta)$, we only need to take the MLE as $\hat{\theta}_n^{MLE} = X_{(1)}$. On the other hand, since $EX = \int_{\theta}^{\infty} \theta x^{-2} \cdot x \, dx = \infty$, the method of moments estimator does not exist.

Problem 2. Let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 \leq x \leq 1, 0 < \theta < \infty.$$

- (1) Find the maximum likelihood estimator $\hat{\theta}_n$ of θ .
- (2) Evaluate $E\hat{\theta}_n$ and $\text{Var}(\hat{\theta}_n)$. Is $\hat{\theta}_n$ unbiased? Is $\hat{\theta}_n$ consistent?

Solution. (1) The log likelihood function is

$$L(\theta) = n \log \theta + (\theta - 1) \sum_i \log X_i.$$

Hence by solving

$$0 = L'(\theta) = \frac{n}{\theta} + \sum_i \log X_i,$$

we have

$$\hat{\theta}_n = \left(-\frac{1}{n} \sum_i \log X_i \right)^{-1}.$$

Since $L''(\hat{\theta}_n) < 0$, $\hat{\theta}_n$ is indeed the MLE.

(2) Since $Y_i \equiv -\log X_i \sim \exp(1/\theta)$ and therefore $-\sum_i \log X_i \sim \text{Gamma}(n, 1/\theta) \equiv T$, it follows that $\hat{\theta}_n =_d n/T$. Hence,

$$E\hat{\theta}_n = nE(1/T) = \frac{n\theta^n}{\Gamma(\theta)} \int_0^{\infty} (1/t) t^{n-1} e^{-\theta t} \, dt = \frac{n}{n-1} \cdot \theta,$$

$$\text{Var}(\hat{\theta}_n) = \dots = \frac{n}{(n-1)(n-2)} \theta^2 \rightarrow 0,$$

as $n \rightarrow \infty$. This shows that $\hat{\theta}_n$ is not unbiased, but is asymptotically consistent.